

LIVER (Surgery)

Anatomy

- RUQ, ~1.5 kg. **Middle hepatic vein divides into right (60%) + left lobes.** Glisson's capsule + peritoneum (except posterior "**bare area**").
- Ligaments: left/right triangular (to diaphragm), **falciform (remnant umbilical vein)**, lesser omentum (hilar structures in free edge).
- **8 segments (Couinaud).** Blood: **portal vein 80% + hepatic artery 20%**; 3 hepatic veins (R/M/L). **LPV longer than RPV.** Common hepatic a. → (GDA) → proper hepatic → R + L. **Cantlie's line** (GB fossa → MHV) divides functional right/left.
- **Only organ that regenerates.** Anhepatic survival 24–48h.

Acute & chronic liver disease

- **Clinical:** malaise, jaundice, confusion/drowsiness, abdominal pain/swelling, N/V, ascites, tremor, bruising, oedema, **fetor hepaticus**. **Most useful sign = flapping tremor (asterixis).**
- **Acute liver failure (ALF):** sudden severe dysfunction + encephalopathy + coagulopathy in previously normal liver. **AASLD: INR >1.5 + any encephalopathy + illness <26 weeks.** Incidence <10/million; **mortality 30–40%**; transplant for some.
- **Chronic liver disease:** 3rd leading premature death cause UK; 90% preventable, 75% present late. Hyperdynamic circulation. **Skin: spider naevi, palmar erythema, leukonychia (white nails); endocrine: hypogonadism, gynaecomastia.** Encephalopathy, ascites, sarcopenia.
- **Prognostic scores: Child-Turcotte-Pugh (CTP) + MELD (INR + bilirubin + creatinine).**
- **Surgical mortality:** CTP-A 10%, CTP-B 30%, **CTP-C 75–80%.** MELD: +1% per point up to 20, +2% above 20.

Investigation

- **US = first-line** for jaundice + RUQ pain (biliary dilatation, not aetiology).
- **CT:** arterial phase detects small **HCC (arterial supply)**; **haemangioma = late venous enhancement**; inflammatory = rim enhancement; stages/resectability (except peritoneal mets).
- **MRI** superior for focal lesions; **MRCP** = biliary tree (≈ direct cholangiography).
- **PET-CT:** functional (FDG); false positive = inflammation.
- **Biopsy of resectable lesions CONTRAINDICATED** → **tumour seeding** (esp. HCC).

Liver trauma

- 2nd most injured abdominal organ (after spleen). Blunt (contusion/laceration/avulsion, ± spleen/renal injury, higher mortality) vs penetrating (± chest).
- **Blood loss = major early complication** (liver gets 20% CO). **FAST** for free fluid.
- **Management depends on haemodynamic stability, NOT injury grade:** unstable → laparotomy; **stable** → **non-operative** (most). Peritonitis → surgery. Conservative success → discharge 8–10 days, rescan 6–8 wks. Operative: packing, suture ligation, resectional debridement, hepatotomy.

Portal hypertension

- Mostly **cirrhosis**; also portal vein occlusion, veno-occlusive, **Budd-Chiari**. Asymptomatic itself → presents decompensated (encephalopathy, ascites, variceal bleed).

- **Variceal bleeding (emergency)**: lower oesophagus commonest; resuscitate (ICU, blood); **vit K + FFP** for coagulopathy; platelets if $<50,000$; **splanchnic vasoconstrictors (terlipressin, octreotide, somatostatin) + prophylactic antibiotics**. Endoscopy confirms (30% non-variceal). **Band ligation** (preferred) or sclerotherapy; balloon tamponade; **TIPSS**.
- **TIPSS**: replaced surgical portocaval shunt; preferred for refractory portal HTN. Complications: liver capsule perforation, **encephalopathy 40%**, stenosis 50% at 1yr. **CI: portal vein occlusion**.
- Surgical shunts (selective/portocaval) — now rare (TIPSS + transplant). **Liver transplant = only treatment of both portal HTN + underlying disease**.
- **Ascites**: **SAAG >1.1 g/dL = portal hypertension**; cytology/culture exclude malignancy/TB.

Specific chronic liver diseases

- **PSC**: associated with **UC**; **young men 30s**; progresses despite colectomy; **↑cholangiocarcinoma + GB cancer risk**; transplant before malignancy.
- **PBC**: **females**; anti-mitochondrial antibodies; malaise/pruritus; transplant definitive.
- **NAFLD/NASH**: **most prevalent chronic liver disease worldwide** (NAFLD 25%, NASH 1.5–6%); 20% NASH → cirrhosis.
- **Budd-Chiari**: hepatic venous outflow obstruction; **young women**; classic triad **abdominal pain + ascites + hepatomegaly**; hypercoagulable in 75% (AT3/protein C/S); **lifelong anticoagulation**.

Liver infections

- **Ascending cholangitis**: biliary infection + obstruction. **Charcot's triad = jaundice + rigors + RUQ tenderness**. Organisms: **E. coli (25–50%), Klebsiella, Enterobacter**. US dilated ducts + obstructive LFT + blood culture. **Broad-spectrum antibiotics + rehydration + urgent drainage (ERCP/sphincterotomy/PTC)**. Commonest cause = stones.
- **Pyogenic liver abscess**: commonest; polymicrobial (**Klebsiella, E. coli, Strep milleri**); elderly/diabetic/immunosuppressed; **biliary source commonest (35%)**, portal spread (20%). Aspiration + C&S; metronidazole/clindamycin + 3rd-gen cephalosporin.
- **Hydatid (Echinococcus)**: **albendazole** before + after surgery.
- **Amoebic abscess (Entamoeba histolytica)**: **"anchovy sauce" pus** (sterile, no odour); **metronidazole + diloxanide furoate**. Drainage indications: **left lobe (pericardium rupture risk), >5 cm/superficial, poor response 3–5 days, diagnostic uncertainty**. Rupture → empyema/peritonitis/tamponade.
- **Fasciola hepatica**: watercress; **triclabendazole**.

Liver tumours

Benign

- **Hepatic adenoma**: **women 25–50, OCP-associated**, solitary; **10% → HCC**; rupture/bleed risk. **Excise if >5 cm** (or stop OCP → may regress).
- **Focal nodular hyperplasia (FNH)**: middle-aged women, no liver disease, **hepatocytes + Kupffer cells**; **NO malignant potential → no treatment**. MRI (liver-specific contrast) distinguishes from mets.
- **Haemangioma**: **most common benign** (5–7%, F:M 5:1), often multiple; incidental; resect only if symptomatic.

Cystic

- **Simple cyst** (5% population, F $>$ M); **polycystic liver** (with adult PKD; **10% cerebral aneurysms**); **cystadenoma** (multilocular, septated; mesenchymal stroma = females, CA19-9 may ↑; premalignant → cystadenocarcinoma).

Malignant

- **HCC**: commonest primary liver cancer; **5th cancer in men**; leading death in cirrhosis; **men 3:1**; 80% Asia/Africa. **Risk**: **HBV (>50%)**, **HCV (17x)**, **aflatoxin**, **alcohol**, **obesity/DM**, **cirrhosis**. Median survival 6–20 mo; <50% at 2yr.
- Staging: **Barcelona (BCLC)**. Only 20–40% resectable. CI: extrahepatic mets, multiple/bilobar, main bile duct/portal vein/IVC thrombus. **Recurrence 80% in 5 yrs**.
- **Transplant = Milan criteria** ($\leq 5\text{cm}$ single or ≤ 3 nodules $\leq 3\text{cm}$; no angioinvasion/extrahepatic) → 75% 4-yr survival.
- If inadequate **future liver remnant (FLR)** → **portal vein embolisation (PVE)**. Resection only CTP-A (minor in B; not C).
- **Intrahepatic cholangiocarcinoma (ICC)**: 20% of cholangiocarcinomas; 2nd to HCC; ↑ in SE Asia (**Clonorchis sinensis**). Subtypes: periductal/mass-forming/intraductal. Aggressive — only 30% resectable; **avoid biopsy**; 5-yr survival 3%.
- **Colorectal liver metastases (CRLM)**: up to 70% of CRC develop mets. **Resection = only curative** (5-yr survival 50%); only 20% candidates (+20–30% after chemo). **R0 margin 1cm gold standard** (R1 similar with modern chemo). **FLR 25% sufficient**. Chemo: **5-FU/folinic + oxaliplatin (50–60%)** ± anti-VEGF/EGFR. **Liver-first approach** for synchronous if liver at risk.

High-yield one-liners

- **Portal vein 80% + hepatic artery 20%**; 8 segments; Cantlie's line; only organ that regenerates.
- **ALF = INR >1.5 + encephalopathy + <26wk**, no prior liver disease; flapping tremor = best sign.
- **CTP + MELD (INR/bilirubin/creatinine)** for prognosis; CTP-C surgery 75–80% mortality.
- **Liver trauma**: management by haemodynamic stability not grade; stable → conservative.
- **Variceal bleed**: terlipressin/octreotide + antibiotics + band ligation → **TIPSS**. SAAG >1.1 = portal HTN.
- **PSC = young men + UC + cholangiocarcinoma**; **PBC = women + AMA**; Budd-Chiari = young women, triad, anticoagulate.
- **Charcot's triad = ascending cholangitis (E. coli)**; amoebic = anchovy sauce pus.
- **Adenoma = OCP**, 10% HCC, excise >5cm; **FNH = no treatment**; haemangioma = commonest benign.
- **HCC**: HBV/HCV/cirrhosis; BCLC staging; Milan criteria for transplant; biopsy contraindicated (seeding).
- **CRC mets**: resection only cure (50% 5yr); FLR 25%; PVE if small remnant.